

CS 486/586

Week 01-1
Databases

Esme Basnet



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What is Data?



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A close-up photograph of a newborn baby's feet being held by an adult's hands. The baby is wearing white hospital socks with identification bands. The adult's hands are gently cupping the feet.



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What is Data?



Is this a **data**?

What is Data?



Did you know your
heart generates data
every second?

What is Data?



What is Data?

Data is information that describes real-world entities, events, or systems.

Examples:



Student grades



Bank transactions

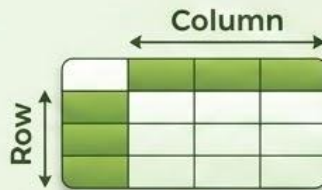


Social media posts



Sensor readings

Types of Data:



Structured Data

Organized in tables
(rows & columns)



Unstructured Data

Images, videos, text



Semi-structured Data

JSON, XML



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Why Data Matters

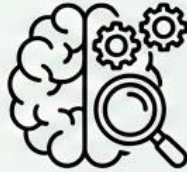


Why Data Matters



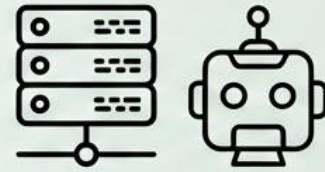
Powers Modern Applications

- Apps, websites, banking, streaming.



Drives Decision-Making & Strategy

- Recommendations (Netflix, Amazon)
- Fraud detection
- Pricing strategies



Foundation of Future Tech

- Software systems
- Cloud computing
- Artificial Intelligence

What is a Database?

A database is an organized collection of related data stored electronically.

Key Characteristics:

- 🗄️ Stores large amounts of data
- ✓ Organized in structured format
- ✓ Supports efficient querying and updates
- ✓ Shared by multiple users and applications

Examples:

- 🗄️ University database (students, courses, grades)
- ✓ Banking database (accounts, transactions)



Why Not Just Use Files?

Problems with File Systems:

-  Data duplication and inconsistency
-  No control over simultaneous access
-  Weak security
-  No validation of incorrect data
-  No recovery from failures

Result:

File systems do not scale for large, complex applications.

Example of Inconsistent data in Excel file

BANK NAMES (Inconsistent)

HSBC
HSBC Bank
HSBC Bk
HSBC Bank (Europe)
Hong Kong Shanghai Bank
HSBC Bank USA
hsbc banking
HSBC Credit Union
NA HSBC Bank & Credit Union
HSBC Mtg Srvs Inc
HBSC Bank
HSBC Corp
hsbc finance cor

COMPANY NAMES (Inconsistent)

GE
General Elec
GEN ELECTRIC
G.E.
GE Corp
General Electric Corporation
GE Corporation
GE Capital

LOCATION/ABBREVIATIONS (Inconsistent)

California -> CA	UK -> UK
CALIF -> CA	United Kingdom -> UK
Cal -> CA	U.K. -> UK
Calif. -> CA	Britain -> UK
Califonia -> CA	Great Britain -> UK
cali -> CA	GR BRITAIN -> UK
	Royaume-Uni -> UK

PERSON NAMES (Inconsistent)

Dr Fred Johnston
Frederick P Johnson
FRED JOHNSON
Freddie Johnson
Sallie Harrell
Sally Harrel
Ms Sally P Harell
Jim Donaven
Mr James Donavon
James Donovan
Gavin McMillan
Gaven P. MacMillen III
E Donnelly
Elizabeth Donnelly
Lisabeth Donley
Mr. Robert J. Ford
Rob Ford
Bob Ford
Robert Forde

Why Do We Use Databases?



Ensure data consistency and accuracy



Provide fast and efficient access



Support multiple users simultaneously



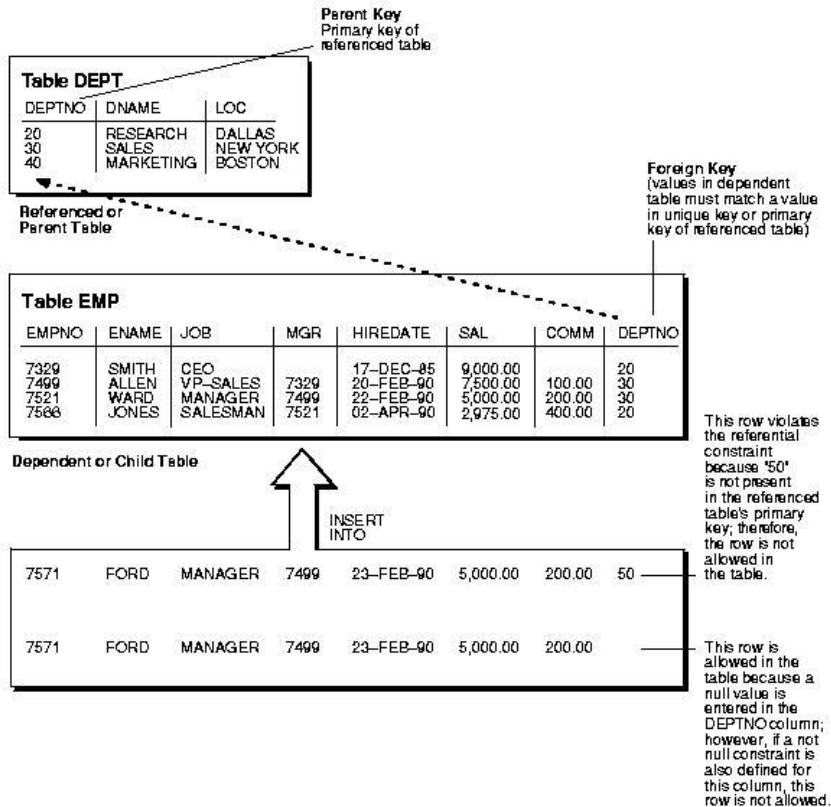
Offer security and access control



Enable backup and recovery



Why Do We Use Databases?



What is a DBMS?

A Database Management System (DBMS) is software that manages databases.

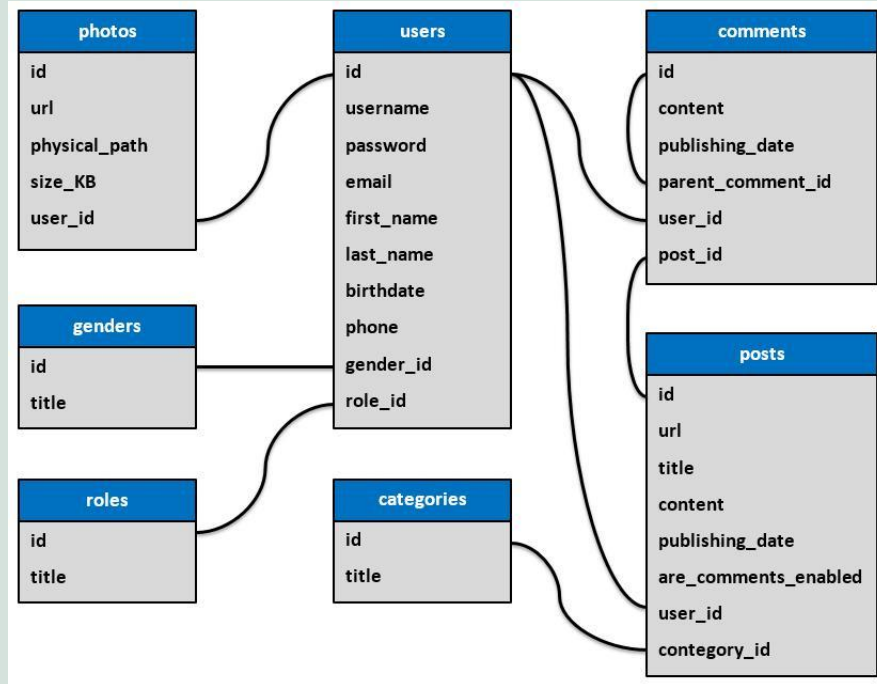
Responsibilities

-  Create and manage databases
-  Store and organize data
-  Allow users and applications to access data
-  Ensure security and reliability

Examples



What is a DBMS?



History of Databases

Pre-1960s:

- Manual records and file-based systems

1960s–1970s:

- Hierarchical databases (tree structure)
- Network databases (more flexible but complex)

1970:

- Relational model introduced

1980s–1990s:

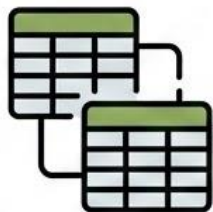
- SQL becomes standard

2000–Present:

- NoSQL databases
- Cloud and distributed databases

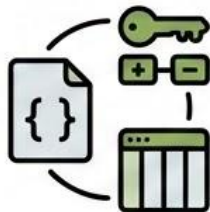
Types of Databases

1. Relational Databases (RDBMS)



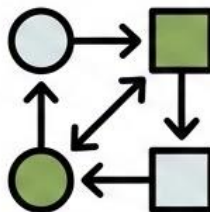
Structured tables
Examples: PostgreSQL,
MySQL

2. NoSQL Databases



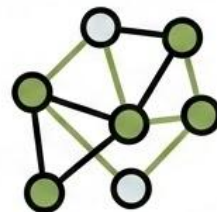
Flexible, large-scale
data
Examples: MongoDB,
Cassandra

3. Object-Oriented Databases



Store objects instead
of tables

4. Graph Databases



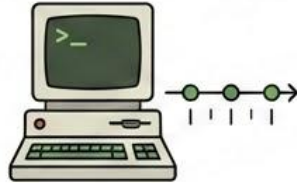
Store relationships
Example: Neo4j

Relational Databases (RDBMS)

Relational Database Management Systems



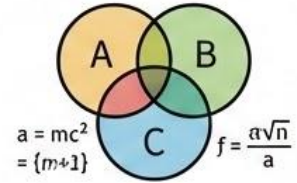
**Fundamental for
storing and
managing data**



Invented ~1970s,
IBM - System R
UC Berkeley
researchers (Ingres)



**Gold Standard,
ubiquitous**



**Formal
foundations in
set theory**

RDBMS - key features



Store data in tables (rows and columns)



Based on structured schema



Use SQL (Structured Query Language)

Key Features:



Schema (defined structure)



Declarative queries



Query optimization



ACID guarantees



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RDBMS - example products



- Oracle RDBMS



- Microsoft SQL Server, Access



- PostgreSQL, MySQL, SQLite, MariaDB



- IBM Db2



- SAP Sybase, Teradata
(and many more)

Tables in Relational Databases

A table is the basic structure of a relational database.



Row (Tuple/Record): A single entry



Column (Attribute): A property of data



Table (Relation): Collection of rows and columns

Relational Databases Consist of **Tables**

Simplified, example **table**
for tracking accounts

Tables sometimes
referred to as **relations**

Account

Number	Owner	Balance	Type
101	Smith	\$1,000	checking
102	Wei	\$2,000	checking
103	Smith	\$5,000	savings
104	Jones	\$1,000	checking
105	Martin	\$10,000	checking

table structure

Table name

column (attribute) names



Account

Number	Owner	Balance	Type
101	Smith	\$1,000	checking
102	Wei	\$2,000	checking
103	Smith	\$5,000	savings
104	Jones	\$1,000	checking
105	Martin	\$10,000	checking

table rows

Each entry in the table is called a **row**.
a.k.a., **tuple** or **record**.

Account



Number	Owner	Balance	Type
101	Ashmy	\$10.00	Decurent
102	Fakferson	\$30.00	Chair
103	John	\$40.00	Uset
104	Helly	\$70.00	Decurent
105	Jamathan	\$50.00	Uset

table instance

instance = the current contents or data in the table



Account	Number	Owner	Balance	Type
	101	Smith	\$1,000	checking
	102	Wei	\$2,000	checking
	103	Smith	\$5,000	savings
	104	Jones	\$1,000	checking
	105	Martin	\$10,000	checking

an instance of the Account table

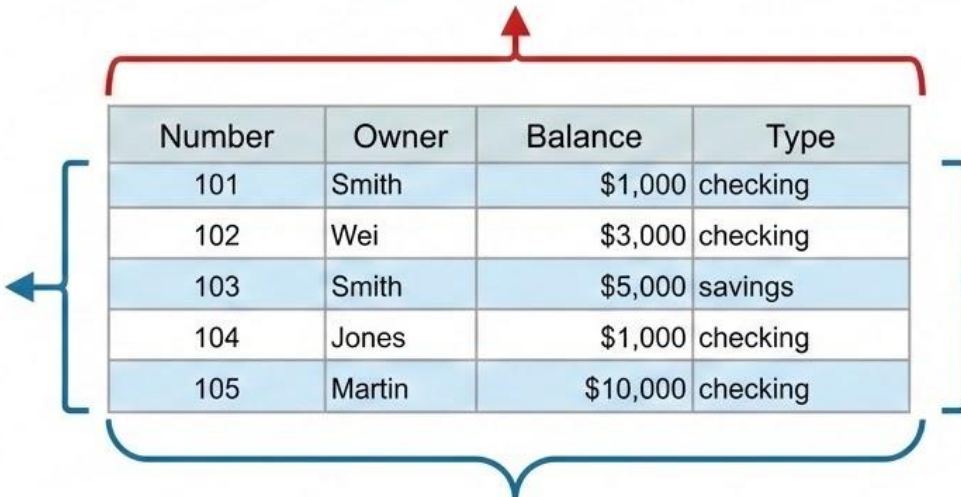
another table instance

another instance of the table
(one row deleted, two rows added)

Account	Number	Owner	Balance	Type
	101	Smith	\$1,000	checking
	102	Wei	\$3,000	checking
	104	Jones	\$1,000	checking
	105	Martin	\$10,000	checking
	107	Yu	\$7,500	savings
	109	Jones	\$450	checking

SIZE OF A TABLE

ARITY = # of columns (4 in this case)



Number	Owner	Balance	Type
101	Smith	\$1,000	checking
102	Wei	\$3,000	checking
103	Smith	\$5,000	savings
104	Jones	\$1,000	checking
105	Martin	\$10,000	checking

CARDINALITY =
of rows
(5 in this case)

DEGREE = another name for **ARITY**

Database = collection of tables

Account



Number	Owner	Balance	Type
101	Smith	\$1,000	checking
102	Wei	\$2,000	checking
103	Smith	\$5,000	savings
104	Jones	\$1,000	checking
105	Martin	\$10,000	checking

Deposit



AcctNo	Deposit-id	Date	Amount
102	1	9/1/2024	\$500
102	2	9/2/2024	\$200
104	3	9/3/2024	\$1,000
105	4	9/4/2024	\$10,000

Debit



AcctNo	Debit-id	Date	Amount
101	924	9/5/2024	\$145
101	925	9/5/2024	\$25

Summary: Relational Structure

Database & Table



Database = collection of tables



Table = collection of columns

- provides structure which enables consistency, performance, security.

Instance & Size



Table Instance = collection of rows

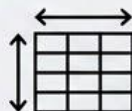


Table Size:

-  **Arity** = # columns
-  **Cardinality** = # rows



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